

Sharon Chepngeno Ngetich

Control and Instrumentation Engineer

Contact

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Profile

Analytical and mission-driven Control & Instrumentation Engineer with strong expertise in maintenance, process automation, and instrumentation systems. Skilled in preventive and corrective maintenance, troubleshooting, and process optimization across manufacturing and energy systems. Adept at transforming operational data into verifiable insights that enhance system reliability and sustainability, a foundation ideal for Monitoring, Reporting & Verification (MRV) and maintenance data frameworks. Currently expanding expertise in AWS Cloud Computing, Embedded system and IIoT for digital monitoring and Control & Instrumentation.

Education

Amazon Web Services (AWS) (Sept 2025 – Dec 2025)

Cloud Computing – Cloud Practitioner Certification

Jomo Kenyatta University of Agriculture & Technology (JKUAT) (Sept 2020–Apr 2024)

BSc. Control & Instrumentation Engineering - Second Class Upper

AIC Litein Girls High School (Feb 2016 – Nov 2019)

Secondary School Education - KCSE B+

Chumo Educational Centre (2012 – 2015)

Primary School Education – KCPE 392/500

Professional Experience

Lead Control and Automation Engineer - Zen Carbon (Apr 2026 – Present) s

- Multidisciplinary engineer specializing in industrial automation, embedded systems, and research & development of scalable engineering solutions.
- Designed, developed, and implemented PLC-based control systems for industrial process automation and monitoring.
- Programmed and configured HMI and SCADA systems for real-time data visualization, alarm management, and process optimization.
- Integrated sensors, actuators, embedded systems, and industrial communication protocols (Modbus RTU/TCP, TCP/IP) for efficient system control.
- Led system prototyping, testing, commissioning, and validation to ensure performance, safety, and reliability.
- Performed electrical panel design, hardware integration, and control logic development for industrial applications.
- Conducted troubleshooting, diagnostics, repair, and preventive maintenance of automation and control systems.

Technical Trainer and R&D –Centurion Systems Limited (CSL) (Feb 2026 – Apr 2026)

- Delivered hands-on training on automation and access control systems to installers, technicians, and clients.
- Developed structured technical training materials, manuals, and practical demonstrations to support learning and system adoption.
- Provided technical support and guidance to installers during system setup, configuration, and troubleshooting.
- Conducted on-site and workshop-based training sessions to ensure proper installation and operation of systems.
- Supported project implementation by offering product expertise and ensuring correct system integration and performance.
- Assisted clients in understanding system functionality, maintenance procedures, and operational best practices.

Technical Sales Engineer and R&D(IIoT) – Centurion Systems Limited (Jan 2026 – April 2026)

- Technical Sales Engineer with a strong background in control systems, automation, and instrumentation, combining hands on engineering expertise with customer-focused solution delivery.
- Experienced in PLC and VFD based automation systems, including system configuration, commissioning, troubleshooting and optimization of industrial processes.
- Skilled in industrial instrumentation covering selection, integration, calibration and validation of pressure, temperature, flow and level measurement systems.
- Proficient in embedded systems development and industrial interfacing, supporting real time control applications and system integration.
- Conduct testing, validation, and performance verification of repaired industrial equipment to ensure safe, reliable, and standards compliant operation.
- Provide end to end project support from pre-sales technical consultation and solution design through commissioning, fault diagnosis, performance testing, documentation and after sales support.
- Contribute to Research and Development (R&D) initiatives in Industrial Internet of Things (IIoT), including data acquisition, remote monitoring, digital integration and implementation of smart automation solutions to improve reliability, reduce downtime and enhance productivity.

Maintenance Technician (Control & Instrumentation) – Glacier Products Ltd (Dairyland) (Sept 2024 – Sept 2025)

- Provided Technical support for PLC and PID controlled production and Refrigeration system and Implemented and troubleshot PID and PLC control systems to enhance process reliability.
- Conducted mechanical, electrical, HVAC and instrumentation maintenance for production and refrigeration systems.
- Prepared Technical Documentation, Maintenance Reports, Work Records & Monitored and analyzed KPIs (MTTR, MTBF, availability) to improve maintenance strategies.
- Ensured compliance with QMS, FSMS, and OSHA standards while optimizing maintenance schedules.
- Worked closely with internal teams, suppliers and service providers to resolve technical issues and Led root cause analysis (RCA) and Operational Excellence (OpEx) initiatives to reduce downtime.
- Prepared and reviewed technical quotations for maintenance spares, instrumentation components, and service support in coordination with suppliers and internal teams.
- Supported cost evaluation and comparison of supplier quotations to ensure technical compliance and value for money.

Industrial Systems Instrumentation & Automation (ISA) Training (Jul 2024 – Sept 2024)

- Programmed Siemens S7-1200 PLCs and HMI using TIA Portal for industrial automation projects.
- Designed and wired control panels integrating controllers, sensors and actuators for real-time monitoring.
- Applied process logic design, P&ID and PFD interpretation and troubleshooting to improve automation system performance.

Instrumentation & Control Attachee – Kenya Electricity Generating Company (KenGen) (May – Jul 2022)

- Calibrated and verified pressure, temperature, and level transmitters using HART communicators and used Meggar meter for insulation test.
- Maintained accurate technical records and store inventory records ensuring traceable data integrity.
- Conducted loop checks, troubleshooting, installation, repair and assisted with instrumentation maintenance in geothermal systems.

Projects

- **Lifecycle Data Monitoring Dashboard:** Built Excel dashboard for maintenance KPIs (MTTR, MTBF).
- **Smart Farm IoT Monitoring System:** Developed ESP32 IoT system using Blynk for remote environmental monitoring.
- **STM32 32-Relay Control System (FreeRTOS + Modbus RTU/TCP):** Designed a 32-relay STM32 controller with Modbus TCP/RTU and FreeRTOS multitasking.
- **Automated Water Transfer System (PLC/HMI):** Programmed Siemens PLC for automated control.
- **PID Integration Using Thermocouple Instead of RTD:** Adapted PID control using thermocouple sensors.
- **AutoCAD Design:** Created optimized chiller and cold room floor plans.

Technical Skills

- **Engineering Fields & Systems:** Control & Instrumentation | Mechanical | Electrical | HVAC
- **Sensors:** Pressure | Temperature | Flow | Level
- **Maintenance & Reliability:** Preventive & Corrective Maintenance | RCA | KPI Tracking | Equipment Diagnostics | MTTR, MTBF & MTTF
- **Automation & Control:** PLC | DCS | SCADA | HMI | PID | VFD
- **Embedded Systems:** STM32 | ESP32 | Raspberry Pi | FreeRTOS | Modbus RTU/TCP
- **IIoT & Cloud:** AWS Cloud Computing | ESP32 | Arduino uno & nano | Blynk | Sensor Networking
- **Reporting & Data Analysis:** Excel | PPT | Analytics | RCA | Tacking & Reporting | Dashboard Design | MRV | Quotation preparation and evaluation | Supplier Coordination & Cost Comparison
- **Compliance:** OSHA | FSMS | QMS | LOTO
- **Programming:** Python | C++ | MATLAB | Bash scripting | Linux | SQL
- **Software:** MATLAB | Simulink | Proteus | TIA Portal | STM32CubeIDE | AutoCAD

Soft Skills

Analytical Thinking | Technical Writing | Collaboration | Problem-Solving | Communication | Time Management | Teamwork | Eager to Learn

References

Available upon requests